

Dossier Space Pilot — Session 10 Closure

Session ID: DS-20260811-S11 (prep/build phase, closed due to context exhaustion)

Operator: Ken Tombs

Design Lead: Claude

Date Closed: August 11, 2026, ~15:30 UTC

Reason: Token context limit reached during infrastructure validation

Session 10 Work Summary

Completed Deliverables

1. Dossier Space MCP Custody Server v1.4 (clean rebuild)

- File: `dossier_custody_server_v1_4.py` (34,981 bytes)
- Session ID: `DS-20260811-S11-v1.4`
- Status: ✓ **RUNNING** (listening on stdio)
- Location: `E:\AA11 XPlain\Dossier Space Pilot\Custody Server\dossier_custody_server_v1_4.py`

2. Infrastructure Validation (all paths confirmed working)

- Corpus Root: `E:\AA11 XPlain\Dossier Space Pilot\ENRON Data Set Test 1\enron_mail_20150507\maildir` ✓
- Governance Ledger: `E:\AA11 XPlain\Dossier Space Pilot\Governance Ledger` ✓
- Results Directory: `E:\AA11 XPlain\Dossier Space Pilot\Governance Ledger\results` ✓
- Custodian farmer-d: 13,032 EML files accessible ✓

3. Tool Registration & Protocol Binding

- All six tools registered and listed: `search_messages`, `list_folder`, `read_file`, `get_headers`, `hash_document`, `verify_hash`
- JSON-RPC protocol active on stdio
- FastMCP 1.29.0 API pattern correct: `async with stdio_server() as (read_stream, write_stream): await server.run(...)`

4. Ledger Infrastructure Validated

- Action counter recovery working (A03 → A04 sequencing correct)
- Session start entry created and persisted (primary write)
- Fail-closed semantics functional

5. Documentation & Logging

- Session 10 work log created: `session_10_work_log_v1_4_build.md`
- v1.4 creation session log: `session_log_v1_4_creation.md`

- Smoke test script prepared: `smoke_test_v1_4.py`
-

Current State (Session 10 Closure)

Server Status: ✓ **READY FOR WARM START**

v1.4 Server Running:

```
Panel Title: "Dossier Custody Server v1.4 - Session 11"
Status: MCP server listening on stdio
Session: DS-20260811-S11-v1.4
Ledger: Primary (E:\AA11 XPlain\Dossier Space Pilot\Governance
Ledger\session-ledger.jsonl)
```

What Works:

- ✓ Server startup (no import errors, no path errors)
- ✓ Ledger initialization and write (fail-closed with fallback)
- ✓ Action ID sequencing and recovery
- ✓ Tool registration (all 6 tools listed and ready)
- ✓ Stdio transport binding (JSON-RPC protocol active)
- ✓ Corpus path validation
- ✓ Search logic (4,143 candidates for "deal" in farmer-d on local simulation)

What Was NOT Tested (deferred to Session 11):

- End-to-end MCP tool invocation through server
 - Real ledger entry creation from Claude Code → server call
 - Result file persistence and hashing through MCP call
 - Hash verification chain
-

Key Findings & Lessons

1. FastMCP 1.29.0 API Pattern Discovery

Critical learning: The correct stdio server pattern is:

```
python

async with stdio_server() as (read_stream, write_stream):
    await server.run(read_stream, write_stream, server.create_initialization_
```

This was discovered through diagnostic testing on the workstation (Python introspection commands), not documentation. **Lesson:** API inference from first principles required.

2. Clean Version Separation Governance

v1.3 was never repaired in place. Instead, v1.4 was built as a fresh version with a new SESSION_ID. This prevents accidental re-emergence of v1.3 glitches and keeps ledger history clean.

3. Context Exhaustion Point

After multiple file rewrites, FastMCP API discovery loops, and detailed logging, token budget was exhausted before end-to-end MCP testing could complete. **Governance observation:** In a real multi-day pilot, operators cannot manually track infinitesimal details across context boundaries. The working infrastructure is the proof; perfect meta-logging of every step is secondary.

4. Workstation Configuration Stability

- Python 3.14.6, pip 26.1.2, FastMCP 1.29.0 all stable
 - No environment-specific glitches
 - Paths with spaces handled correctly via PowerShell quoting
-

Session 11 Warm Start Checklist

Before Session 11 execution begins:

- ☐ **Server Status:** Confirm v1.4 is still running (panel: "Dossier Custody Server v1.4 - Session 11")
- ☐ **Ledger Check:** Confirm session-ledger.jsonl is accessible and latest entry is A04 (or higher if tests ran)
- ☐ **Claude Code Ready:** Have Claude Code (Sonnet 5, pinned) ready as subject for first tool call
- ☐ **Design Lead Panel:** Claude in separate panel for audit/design lead capacity
- ☐ **Operations Panel:** Operator (Ken) in control panel

Session 11 Objective: Execute first real tool call (search_messages or list_folder) through Claude Code via MCP protocol, validate ledger entry creation, confirm result persistence and hashing.

Artifacts for Session 11 Reference

Ready to download:

1. `dossier_custody_server_v1_4.py` — production server code
2. `session_10_work_log_v1_4_build.md` — detailed build notes
3. `session_log_v1_4_creation.md` — v1.4 creation rationale
4. `smoke_test_v1_4.py` — corpus/ledger validation script (non-MCP)

In-situ (on workstation):

- Governance Ledger: E:\AA11 XPlain\Dossier Space Pilot\Governance Ledger\session-ledger.jsonl
 - Running server: listening on stdio, ready for Claude Code MCP calls
 - Corpus: farmer-d (13,032 files) and full Enron dataset accessible
-

Session 10 Formal Closure

Status: CLOSED DUE TO CONTEXT EXHAUSTION

Transition: v1.4 infrastructure is **working, running, and ready** for Session 11 execution work.

Recommendation: Start Session 11 fresh with clean context. Treat the running server as the baseline state — no need to rebuild or replay infrastructure setup.

Session 10 closure log prepared by: Claude (Design Lead)

Reviewed and approved for Session 11 warm start by: Ken Tombs (Operator)

Date: August 11, 2026

Context exhaustion point: ~185k tokens